Supplementary Material
Where Does Politeness Live in Hindi?
A Mechanistic Case Study of Honorific Encoding in Gemma Scope SAEs

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This supplement expands the experimental and evaluation details reported in the main paper. It is deliberately evidence-bounded: every number and claim below is already stated in the manuscript. The supplement provides a compact audit trail from research question to measurement, control, result, and interpretive limit.

A1 Updated Pre-Marker Evidence Map

This supplement records the additional controls that address the principal surface-morphology alternative: a high probe score might merely detect an overt pronoun or verb-agreement suffix. In the primary context-only, pre-marker condition, the model receives only social and discourse context and the state is extracted before either target form is realized.

Evaluation	Result	Interpretation
Character n-gram baseline	AUROC 0.550	Best neural probe is $+0.172$ higher
Word TF–IDF baseline	AUROC 0.610	Strongest surface baseline; neural gain is $+0.112$
Layer-0 probe	AUROC 0.535	Best neural probe is $+0.187$ higher
Middle-layer probe	AUROC 0.722	Contextual signal beyond both explicit markers
Held-out addressee role	AUROC 0.692	Generalises across the tested role split
Naturalistic pre-marker prefix	AUROC 0.655	Signal is present before an overt marker
f_{646} audit profile	$d = 2.209$; $A_{\text{hon}} = 0.514$; $A_{\text{pron}} = 0.808$	Fails the prespecified specificity criterion
Causal pilot	$n = 5$ prompts; best shift $+0.028$ at $\alpha = 8$	Non-monotonic; no causal control established
Human evaluation	3 native Hindi raters; 192 generations; 576 generation–rater evaluations	Corroborates the bounded negative intervention result

Table A1: Updated evidence map. Gains are AUROC-point differences from the best middle-layer probe (0.722), with the comparison baseline named explicitly.

Interpretive boundary. The pre-marker context-only control rules out an explanation based only on explicit pronouns or agreement morphology in the target utterance. It does not rule out every contextual correlate of social role, topic, or template construction. Accordingly, the paper claims contextual decodability and a scenario-bounded causal pilot, not a complete or uniquely localized honorific mechanism.

A2 Context-Only Dataset and Probe Protocol

A2.1 Pre-marker construction

The context-only evaluation asks a different question from the completed minimal-pair probe. Each item supplies the speaker–addressee relationship, familiarity, communicative setting, and intended propositional content, but ends before the response realizes an *aap*, *tum*, or *tu* form or an agreement-bearing target verb. The label is the honorific level judged appropriate for the next utterance. Residual-stream activations are extracted at the final context token, immediately before generation begins.

This design prevents the classifier from observing the target pronoun or its agreeing morphology. It is not a claim that the input contains no information about social register: social roles, lexical context, and discourse structure can legitimately carry information relevant to the appropriate form of address. The evaluation instead distinguishes prediction of intended register from post-hoc recovery of a marker already present in the string.

Component	Specification
Input	Social and discourse context ending before the target honorific form
Label	Honorific level appropriate for the next utterance
Extraction point	Final context token, before generation begins
Primary split	Grouped cross-validation, with paired scenarios kept together
Generalisation split	Addressee-role family held out of the training fold
Naturalistic check	Prefix from realistic dialogue ending before the first overt honorific marker
Surface controls	Character n-gram and word TF–IDF logistic regression on the same input and splits

Table A2: Context-only, pre-marker protocol. The goal is to predict an intended register before the corresponding overt morphology occurs.

A2.2 Probe and baseline comparators

Completed-sentence probes use mean-pooled residual-stream activations. Context-only pre-marker probes use the residual-stream activation at the final context token. The direct-text baselines use the same classifier family, so the comparison isolates representation type rather than a classifier capacity change. Layer 0 provides a neural input-level reference: improvement at later layers is evidence that the relevant label is more linearly accessible after transformer computation, though it does not by itself localise the computation or establish causal use.

Comparator	AUROC	Difference from 0.722
Layer-0 representation	0.535	-0.187
Character n-grams	0.550	-0.172
Word TF–IDF	0.610	-0.112
Best middle transformer layer	**0.722**	–

Table A3: Context-only probe comparison. Word TF–IDF is the strongest evaluated surface-form baseline; all reported gains in the main paper name the baseline to which they refer.

A3 Generalisation Results and Interpretation

The best middle-layer probe reaches AUROC 0.722, compared with 0.535 at layer 0, 0.550 for character n-grams, and 0.610 for word TF–IDF. It therefore improves by 0.187, 0.172, and 0.112 AUROC points over those comparators, respectively. The gain over the highest-performing surface baseline is the most conservative direct-text comparison.

Two additional evaluations test whether the result depends entirely on the observed scenario construction. Holding out an addressee-role family yields AUROC 0.692; naturalistic dialogue prefixes ending before the first overt marker yield 0.655. The lower values are expected under distribution shift, but remain above the direct-text baselines reported for the primary context-only condition. These results support contextual decodability across the tested splits. They do not rule out residual lexical or social-role correlations, and they should not be read as open-domain performance estimates.

Evaluation	AUROC	What it tests
Primary context-only middle-layer probe	0.722	Contextual prediction before overt marking
Held-out addressee-role probe	0.692	Transfer to unseen role families
Naturalistic pre-marker prefix probe	0.655	Transfer to realistic dialogue context

Table A4: Generalisation checks for the context-only signal.

A4 SAE Audit and Exploratory Causal Pilot

Candidate SAE features are ranked by raw-condition Cohen's d and then audited with two separate, orientation-invariant AUC quantities. A_{hon} measures honorific discrimination on pronoun-stripped pairs, while A_{pron} measures raw-versus-stripped sentence discrimination. The prespecified passing rule is $A_{\text{hon}} \geq 0.80$ and $A_{\text{pron}} \leq 0.55$. This audit tests the raw ranking procedure; it is not an exhaustive context-only search over the SAE dictionary.

Candidate profile	d	A_{hon}	A_{pron}
f_{646} (raw rank 1)	2.209	0.514	0.808
All 50 candidates	–	≤ 0.518	0.656–0.808
Passing rule	–	≥ 0.80	≤ 0.55

Table A5: Prespecified two-way SAE specificity audit. No raw-ranked candidate passes; the top candidate is strongly associated with pronoun presence.

The corrected pair is reported for all 50 candidates. None satisfies $A_{\text{hon}} \geq 0.80$ and $A_{\text{pron}} \leq 0.55$.

The exploratory intervention pilot uses f_{646} at layer 2, five informal prompts, one generation per prompt and dose, and $\alpha \in \{2, 4, 8, 16\}$. The proxy is bounded approximately in $[-1, 1]$; therefore no score of 2 is reported. The baseline mean is -0.100; the corresponding steering means are -0.091, -0.091, -0.072, and -0.216. The largest pre-collapse shift is $+0.028$ at $\alpha = 8$, and the response reverses at $\alpha = 16$. Negative-direction and encode-side zero ablation on five formal prompts preserve positive marking. The pilot consequently does not establish sufficiency or necessity, and its sample is too small for stable confidence intervals or inferential tests.

Condition ($n = 5$ **prompts**)	**Mean proxy**	Δ
Baseline (informal)	-0.100	–
Steering, $\alpha = 2$	-0.091	$+0.009$
Steering, $\alpha = 4$	-0.091	$+0.009$
Steering, $\alpha = 8$	-0.072	$+0.028$
Steering, $\alpha = 16$	-0.216	-0.116
Negative ablation (formal)	$+0.216$	no reduction
Zero ablation (formal)	$+0.245$	no reduction

Table A6: Exploratory causal pilot: feature f_{646}, layer 2, unit-normalised direction, deterministic one-generation decoding.

A5 Human Evaluation and Remaining Limits

Three native Hindi speakers, blinded to condition, independently rated 192 generations spanning eight conditions and 24 matched prompts on six 0–3 axes. This yielded 576 generation–rater evaluations and 3,456 scalar judgments. Mean pairwise Spearman ρ was 0.79 for addressee honorificity, 0.72 for verb agreement, and 0.70 for register consistency. Prompting achieved the highest addressee-honorificity score (2.36 vs. 0.47 for the baseline), whereas top-20 steering reached 1.31 but reduced verb agreement (2.01 vs. 2.68). Thus, the human ratings corroborate the bounded negative result for the tested additive interventions while providing pragmatic checks beyond the automatic morphology proxy.

The strongest remaining alternative explanation is contextual rather than morphological: models and baselines may exploit lexicalised social roles, topics, or scenario regularities. We mitigate this through the held-out-role split, direct-text baselines, and naturalistic prefixes, but broader role reversals, independently validated contexts, and larger open-domain dialogue evaluations remain important future tests. The next confirmatory test is to rank candidates directly on pronoun-stripped or pre-marker contexts, evaluate candidate selection on held-out examples, and then test steering and ablation.

A6 Additional Results: Cross-Model Probe and Per-Condition Ratings

Cross-model probe replication. The completed-sentence probe was replicated on nine models. This is a probe-only check: SAE and causal conclusions remain restricted to Gemma Scope and Gemma-2-2B.

		Raw AUROC		Stripped AUROC	
Model	**Series**	Max	Spread	Max	Spread
Gemma-2-2B	Gemma-2	0.894	0.003	0.817	0.020
Qwen2.5-1.5B	Qwen2.5	0.895	0.003	0.816	0.021
Qwen2.5-3B	Qwen2.5	0.894	0.002	0.817	0.021
Qwen3-4B	Qwen3	0.894	0.002	0.818	0.026
Qwen3-8B	Qwen3	0.893	0.002	0.816	0.019
Llama-3.1-8B	Llama-3	0.893	0.002	0.820	0.018
Nemotron-8B†	Nemotron	0.893	0.002	0.819	0.039
Phi-4-14B	Phi	0.893	0.004	0.816	0.023
Mistral-22B	Mistral	0.892	0.002	0.816	0.009

Table A7: The decodable-but-flat signature replicates across seven model series (Gemma-2, Qwen2.5, Qwen3, Llama-3.1, Nemotron, Phi, Mistral) in nine models spanning six parameter scales (1.5B–22B). "Spread" is the AUROC range over all layers; near-zero raw spread (≤ 0.004) everywhere shows that linear separability varies little with depth. It does not establish where the information is computed or exclude additional representations. †Nemotron-8B is an RLHF fine-tune of Llama-3.1; its higher stripped spread (0.039 vs. ≤ 0.026 for base models) does not affect the raw-curve conclusion.

Per-condition human ratings. Table A8 gives mean ratings for the three honorific-specific axes in all eight conditions, from the 576 generation–rater evaluations described above.

Condition	**Addressee honorificity**	**Verb agreement**	**Register consistency**
4-shot prompting	2.36 ± 0.68	2.74 ± 0.44	2.81 ± 0.40
Baseline (*tu*-form)	0.47 ± 0.50	2.68 ± 0.47	2.61 ± 0.49
Zero-shot instruction	0.19 ± 0.40	2.51 ± 0.58	2.68 ± 0.53
Single SAE feature	0.61 ± 0.62	2.62 ± 0.49	2.78 ± 0.42
Top-5 SAE features	1.01 ± 0.72	1.99 ± 0.93	2.12 ± 0.85
Top-20 SAE features	1.31 ± 0.64	2.01 ± 0.66	2.06 ± 0.84
Supervised diff.-in-means	0.49 ± 0.63	2.68 ± 0.47	2.76 ± 0.46
Random steering	0.39 ± 0.62	2.69 ± 0.46	2.65 ± 0.48

Table A8: Human ratings for all eight conditions (mean $\pm$ standard deviation).